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Population model

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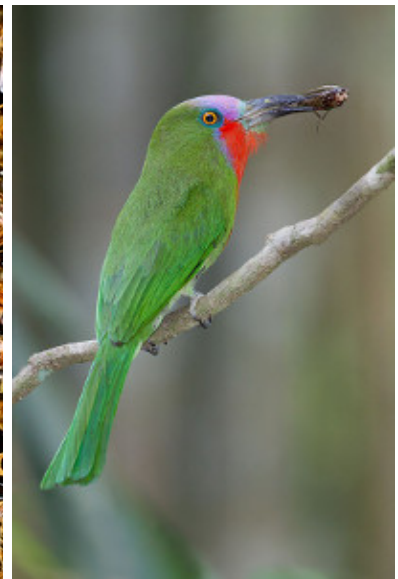


Photo of red-bearded bee-eater by JJ Harrison.

Interacting populations

Choose your own set of populations: animals, plants, bacteria... and model the size of the populations. One population might feed on the other, as with the gouramis and rainbowfish we use in the course in Module 3, but you can also choose two populations that compete for the same resources, such as rabbits and deer competing for the same grass. A third option is that the two species benefit from each other such as cleaner fishes and groupers.

Population model

The number of members in the first population is called $u(t)$ and the number of members of the second population $v(t)$.

$$\begin{aligned}\frac{du}{dt} &= \alpha \left(1 - \frac{u(t)}{k} \pm \frac{\beta v(t)}{1 + \gamma u(t)} \right) u(t) \\ \frac{dv}{dt} &= \left(-\delta \pm \frac{\beta u(t)}{1 + \gamma u(t)} \right) v(t)\end{aligned}$$

The parameters α , k , β , γ , and δ you have to choose appropriately, as well as the signs some of the terms.

Simplified population model

After Module 3, you will let the populations interact, but for now, investigate the case of no interaction: $\beta = 0$.

$$\frac{du}{dt} = \alpha \left(1 - \frac{u(t)}{k} \right) u(t) \quad \text{and separately:} \quad \frac{dv}{dt} = -\delta v(t).$$

Tasks for the project in Module 3

1. Keep trying to collect quantitative information on your populations. If you cannot find that information, make an educated guess. Do not forget to write down where you found the information!
2. Throughout, record your findings. Write down the values you used, store the graphs with descriptive names and write down all conclusions you reached. Store the different versions of the program with descriptive names as well, so you can easily rerun a program later.
3. Choose initial conditions $v(0)$ and $u(0)$.
4. Implement Euler's method for the differential equation for v of the simplified model. Simulate $v(t)$ with different stepsizes. Estimate the error, using the result with the doubled stepsize. Compare the solution with the analytical solution and evaluate how good the error estimate actually is. Decide how precise you want to calculate $v(t)$, and choose a stepsize accordingly.
5. Implement Euler's method for the differential equation for u of the simplified model. Decide on the range of values for α that you want to test. Use the highest value of that range for α to do the stepsize experiments with. (Why the highest value?) Decide how precise you want to calculate u and choose a stepsize accordingly (for this high value of α).

6. Run some experiments for different values of α . Try out how sensitive the results are to changes in α . Choose a value for α that fits the quantitative information you have, or that gives results that seem reasonable. Do not forget to use the unit for α when you write it down.

When you have completed the theoretical part of module 3, you can start already on the full basic model. You can also choose to leave the next points for module 4.

7. Implement Euler's method for the system, with $\beta > 0$, and start with $\gamma = 0$. If you have decided on different stepsizes for the calculations of the separate $v(t)$ and $u(t)$, for the system you need to calculate with the minimum of the two stepsizes. If in doubt check the errors again.

8. Find a reasonable value for β . There are different ways for finding a value:

- A quantitative argument, such as we gave for the gouramis.
- Numerical simulation experiments and see whether the resulting interaction is plausible.
- Analytical calculation of the equilibrium points, especially the one with both populations present. Are the numbers of the populations in that equilibrium in the right proportion? It is probably wise, to start with $\gamma = 0$.
- Calculation of the eigenvalues of the Jacobian matrix in the end state equilibrium point. If the solution oscillates, determine the period of the oscillations and set β to a value such that the period of the oscillations is plausible.

Ideally, you use more methods to determine β . On the other hand, maybe you want to press on, and focus on other parts of the modelling process. Then you use just one method.

9. Start including $\gamma > 0$. Same options to choose a value as for β .

We say it again: do communicate well with your teammate, so you make the different choices together. Document every result and decision in your logbook. That will help in the communication between you two, and when you are writing the report later on, you can find all your previous results easily.

After the pages for the other problems, you will find a page for questions you might have about the project.



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